- The filtrate is treated with n-butanol or methylisobutylketone in acidic or alkaline condition for extracting the antibiotic.
- The process of adsorption to activated charcoal is done to remove other impurities.
- After the elution of chlortetracycline, it is crystallized.

3.2 PRODUCTION OF VITAMINS:

3.2.1 VITAMIN B2:

- Vitamin B2 has been commercially produced by fermentation or chemical synthesis or a combination of both. Fermentation is the most recent and cost effective method.
- However, manufacturers commonly use the mixed process. In this process, a four step reaction sequence is used, starting from glucose. First, ribose is produced from glucose by fermentation. Then a reaction with xylidine is used to convert ribose into riboside, which is then hydrogenated to produce ribamine, which is purified by crystallization.
- Subsequently, a reaction between ribamine and phenyl diazonium salt derived from aniline is started to produce phenylazoribitylamine. This compound is then crystallized, dried and converted into vitamin B2 by cyclocondensation with barbituric acid.

Glucose

↓ Fermentation

Ribose

↓ Xylidine

Ribamine

↓ Aniline

Phenylazoribitylamine

↓ Barbituric acid

Raw riboflavin

↓

Riboflavin (96% pure)

Mixed Process

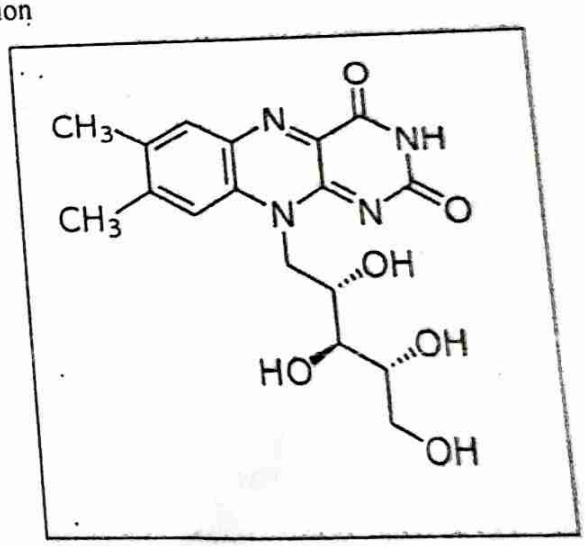
Glucose, Molasses

↓ Fermentation

Riboflavin

↓

Riboflavin (80% Pure)



Fermentation Process

STRAINS OF MICROORGANISMS

- Many strains of bacteria, fungi or yeasts can be used; each manufacturer may be u different organism.
- At present, the bacterium *Bacillus subtilis*, the fungi *Eremothecium ashbyii* and *Ash gossypii* and the yeasts *Candida flareri* and *Saccharomyces cerevisiae* are commonly u in the production of Vitamin B2.
- Kuhn, Gyorgy and Wagner Jauregg in 1933 isolated riboflavin from whey of milk wh it is present in free riboflavin form.
- According to IUPAC rules, riboflavin is called as, 7,8-dimethyl-10-(d ribityl)isoalloxazine, also known as lactoflavin.
- Daily human demand is 1.7 mg. vitamin B2 is cannot store in the body and a consta intake is required.

RAW MATERIALS

- Producers use different basic raw materials. Usually, vegetable oil, glucose or molass corn steep liquor and meat scraps are used as the sources of carbon. However, they va greatly in price and carbon content.
- Further, enzymatically degraded collagen and lipids are used as energy sources togeth with cornsteep liquor.
- The addition of peptones would give better results with bacterial and yeast strains.
- Basic medium consists of corn steep liquor 2.25%, commercial peptone 3.5%, soyabea oil 4.5% but it can be supplemented further by addition of different peptones, glycin distiller's soluble, or yeast extracts.

FERMENTATION:

- The riboflavin fermentations utilize media containing a simplified sugar, such as glucos plus additional crude organic nutrients.
- In certain cases, glucose is totally replaced by corn oil to stimulate riboflavin yields.
- The medium is initially adjusted to pH 6 to 7.5 and the ***Ashbya gossypii*** fermentation is conducted at a temperature of 26 to 28^{0}C for approximately 4 to 5 days.
- Optimal fermentation conditions include aeration at about one third volume of liquid pe minute and agitation.
- Silicon and soyabean oil are commonly used as foam control agents.

- It is a submerged, aerated fermentation, but high level aeration must be avoided, because excess air inhibits mycelial production and reduces the riboflavin yields. Yield is 27 g/l.
- At harvest, the culture is evaporated and dried to serve as a feed supplement.
- Those riboflavin fermentations utilizing species of candida are extremely sensitive to the presence of traces of iron and as a result, iron or steel fermentation equipment cannot be used.
- In fact, it has been proposed that the fermentation equipment be lined with plastic.
- The candida fermentation does have one advantage, however, in that it is carried out at relatively low pH values which inhibit bacterial growth and negate a requirement for sterilization of the fermentation medium.

The fermentation progresses through three phases:

- First Phase- In this phase rapid growth occurs with small quantity of riboflavin production. The utilization of glucose occurs resulting in to decrease in pH due to accumulation of pyruvate. By the end of this phase, the glucose is exhausted and growth ceases.
- Second Phase- Sporulation occurs in this phase. The pyruvate decreases in concentration. Ammonia accumulates because of an increase in deaminase activity. The pH reaches towards alkalinity.
- Third Phase- There is a rapid synthesis of cell bound riboflavin (FMN and FAD). This phase is accompanied by rapid increase in catalase activity.
- As the fermentation completes, the autolysis takes place with releases free riboflavin in to the medium as well as retained in the nucleotide form.
- The riboflavin is present both in solution and bound to the mycelium in the fermentation broth.
- The bound vitamin is released from the cells by heat treatment (1hr, 120°C) and the mycelium is separated and discarded.
- The riboflavin is then further purified.
- Deficiency results in skin disorders.

5,6 dimethylbenzi

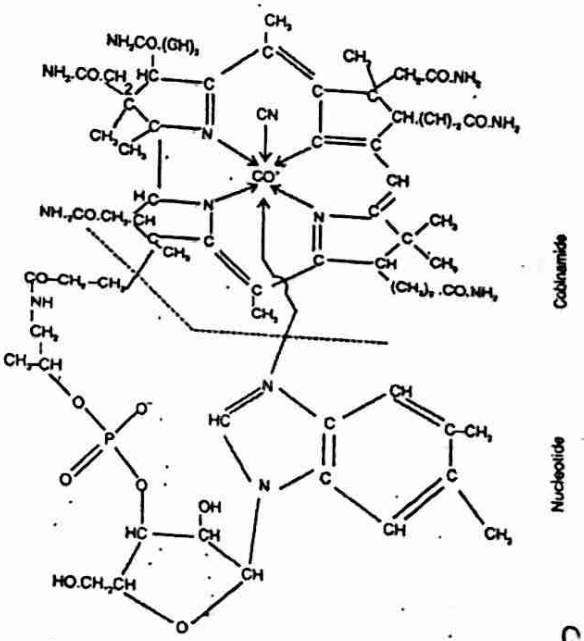
3.2.2 PRODUCTION OF VITAMIN B_{12} ((5,6-dimethylbenzimidazolyl)cobaidcyanide):

- Vitamin B12 is an important dietary component for normal growth in human beings and domesticated animals. Its daily requirement for human beings is 0.001mg/day.

- In nature, vitamin B_{12} is synthesized by microorganisms. For industrial production vitamin B_{12} a number of bacteria and streptomycetes are used.
- The amount of vitamin B_{12} produced by them has been estimated about 20 mg/liter.
- Vitamin B_{12} was first isolated in 1948 by American chemist Karl Folkers and Briti chemist Baron Alexander Todd.

Chemical Structure = $C_{63}H_{88}CoN_{14}O_{14}P$

- Vitamin B_{12} (cyanocobalamin) contains a molecule of cobinamide linked to a nucleotic which has 5,6-dimethyl benziminazole as its base, instead of a purine or pyrimidine bas (Fig.).
- The cobinamide molecule has a central atom of cobalt linked to a cyanide group an surrounded by four reduced pyrol rings joined to form a macroring.



THE MICROORGANISMS:

- *Streptomyces griseus*
- *Streptomyces olivaceus*
- *Bacillus megaterium*
- *Bacillus coagulans*
- *Pseudomonas denitrificans* etc

- Generally VitaminB12 is produced by the submerged (continuous) fermentation process.
- Aeration and agitation of the production medium are essential.
- Usually, such fermentations are completed in 3-5 days.

Vitamin B12 production using *Streptomyces olivaceus*:

Preparation of Inoculum:

- In the building up of the inoculum, pure agar slant culture of *S.olivaceus* NRRL B-1125 is inoculated and grown in 100-250ml of inoculum medium contained in Erlenmeyer flasks.
- These selected flasks are to be kept on the platform of a mechanical shaker during incubation in order to aerate the medium.

Production medium:

- Production media used in this fermentation, usually, consists of carbohydrate, proteinaceous material and a source of cobalt and other salts. A typical production medium is presented as below:

Distillers solubles	4%
Dextrose	0.5-1%
$CaCO_3$	0.5%
$COCl_2.6H_2O$	1.5-10 ppm
pH	7.0 (adjusted with NaOH)

- Most forms of distillers' solubles, soyabean meal, yeast, casein, etc are found satisfactory under suitable conditions.
- It is necessary to add cobalt to the medium for the maximum yields of cobalamin.
- Sterilization of the medium may be practiced batch wise or continuously. In batchwise sterilization, the medium contained in the production tank is heated at 250^0F (120 °C) for 1 hour.

Commercial Production

- Commercially vitamin B_{12} is produced in a continuous culture, where two fermenters are used in a series.

- In each fermenter culture is kept for about 60 hours.
- Precaution taken is that the first fermenter should be anaerobic, while the second aerobic one.
- In the second fermenter 5,6-dimethyl benziminazol is added continuously.
- Sterilized culture medium, containing glucose, corn steep, betain (5%), cobalt (5 ppm) and pH 7.5 in fermenter, is inoculated with *Propionibacterium freudenreichii* and allowed to anaerobic fermentation for about 70 hours.
- During this period cobinamide is produced which accumulates in the broth. Thereafter, 5,6-dimethyl benziminazole (0.1%) is added to it.
- The fermenter is then kept further for 50 hours for aerobic fermentation. During this period nucleotide is synthesized and linked with cobinamide molecule to yield about 20 ppm cobalamin.

Temperature:

- A temperature of 80^0F (26.67 °C) in the production tanks is satisfactory during fermentation.

pH:

- There are a few tenths of a unit fall in pH and a rapid consumption of sugar in the first 24 hours of the fermentation. After 2-4 days, lysis of the mycelium begins resulting in the rise of the pH.
- The stabilization of the mash is practiced by reducing the pH to about 5 with sulfuric acid and adding small amounts of a reducing agent, sodium sulphate.

Aeration and agitation:

- A proper rate of aeration and a current speed of agitation are essential. Aeration rates higher than optimum cause excessive foaming.
- The optimum rate of aeration is about 0.5 volume air/ volume medium/ minute.
- The sterilization of air is, usually, practiced by passing it through columns filled with activated charcoal.

Antifoam agents:

- With this fermentation, foam formation is a serious problem, particularly at the beginning and at the end of the fermentation. There are many antifoam agents that may be used to suppress the formation of foam.

- Important defoaming agents are soyabean oil and corn oil.

Yields:

- The yields of cobalamin are usually in the range of 1-2 mg per liter in the fermented broth.
- Culture is acidified to pH 2.0 to 3.0, gently heated to 100°C and filtered to remove cell debris.
- Finally potassium cyanide (5ppm) is added to filtrate to give cyanocobalamin.
- Using genetic engineering techniques, it has become possible to obtain mutants of micro-organisms, producing more vitamins than the natural ones.
- *P. denitrificans* is able to produce 50,000 times more vitamin B_{12} than its parental strain.

Recovery:

- Heating the mixture to boiling at pH 5 or below liberates the cobalamin quantitatively from the mycelium.
- In order to obtain crystalline vitamin B12, the very first operation is of filtration to remove mycelium.
- Then the filtered broth is treated with cyanide to bring about the conversion of cobalamin to cyanocobalamin.
- The adsorption of the cyanocobalamin from the solution is practiced by passing it through an adsorbing agent packed in columns. Several adsorbents are available; activated charcoal, bentonite, fuller's earth and ion exchange resins.
- Finally, elution of cyanocobalamin from adsorbent is accomplished by the use of an aqueous solution of materials ranging from organic basis to HCl. Several materials have been reported: water, water-acetone and solutions of sodiumcyanide or sodium thiocyanate.

Health functions:

- A sufficient intake of vitamin B12, also known as cobalamin, is important as it helps the body to:
  - Convert food into glucose, which is used to produce energy
  - Maintain healthy nerve cells
  - Produce nucleic acids (DNA), the body's genetic material

- Regulate, together with vitamin B9 (folate), the formation of red blood cells
- Control, together with vitamin B6 and vitamin B9, blood levels of the amino ac homocysteine, a potential marker for heart disease risk.

- Low levels of B12 can cause a range of symptoms including fatigue, shortness of breat diarrhea, nervousness, numbness, or a tingling sensation in the fingers and toes.
- Severe deficiency of B12 causes neurological damage.

3.2.3 PRODUCTION OF VITAMIN A (RETINOL, RETINAL= $C_{20}H_{30}O$):

- Vitamin A is a fat-soluble vitamin.
- Two different types of vitamin A are found in the diet.
- Preformed vitamin A is found in animal products such as meat, fish, poultry and dairy foods.
- The other type, pro-vitamin A is found in plant-based foods such as fruits and vegetables. The most common type of pro-vitamin A is beta-carotene.
- Vitamin A is also available in dietary supplements, usually in the form of retinyl acetate or retinyl palmitate (preformed vitamin A), beta-carotene (pro-vitamin A) or a combination of preformed and pro-vitamin A.
- β-Carotene, also known as provitamin A occurs naturally as a component of various agricultural products and particularly as a component of green plants. When ingested, it is converted into vitamin A in the animal body. Although β-Carotene can be produced by fermentation, the yields at present time are relatively very low.
- β-Carotene is produced by various organisms, but particularly by members of the Choanephoreaceae family of the phycomycetes. Thus, *phycomyces blakesleeanus*, *choanephora cucurbitarum* and *Blakeslea trispora* have received extensive study for their ability to produce β-Carotene.
- The fermentation medium among various ingredients includes β-ionone, nonionic detergent and vegetable oils, fats, fatty acids, waxes or white grease.
- The β-ionone, although it would appear to be a possible precursor of β-carotene, apparently is not directly incorporated into the β-Carotene molecule. Instead, it is regarded as a "steering" factor associated with bringing about the presence and activity of the enzymes responsible for β-Carotene formation.
- Inoculum of each mating type (Phycomyces blakesleenus / Blakeslea trispora) is grown separately and approximately 5 percent of each is added to the fermentation medium.

although it is not necessary that both mating types be added to the fermentation approximately 48 hours after inoculation.

3.2.4 PRODUCTION OF VITAMIN C (ascorbic acid):

Properties of vitamin C:

- Ascorbic acid (vitamin C) is an essential organic compound required for many biosynthetic activities in most of the animals and all humans.
- Ascorbic acid, the official name of vitamin C, was first discovered in 1912. In 1933 it was isolated for the first time in lemons.
- It appears as white or light yellow crystals or powder.
- Its empirical formula is $C_6H_8O_6$. It is made up of elements carbon, hydrogen and oxygen.
- It is water soluble. In alcohol it dissolves sparingly. In benzene, ether and chloroform it is insoluble.
- It is also known as L-ascorbate. At the temperature of 190-192°C it decomposes.
- Its bioavailability is rapid and complete.
- It binds negligibly with proteins. Its metabolic half life is 30 minutes.
- It is excreted by the kidneys.

Biosynthesis of vitamin C in animals and the human inability:

- It is biosynthesized by reptiles and birds in their kidneys. Birds of recent orders and most of the mammals synthesize it in their livers.
- In these animals glucose is converted into ascorbic acid with the help of the enzyme L-gulonolactone oxidase.
- In most of these animals ascorbic acid is a natural metabolite of liver.
- Humans, most of the primates and guinea pigs cannot produce vitamin C in the body and they are required to source it from their food and can develop its deficiency.

Commercial production of Ascorbic acid:

- Earlier vitamin C has been made commercially by extracting it from plants. Later vitamin C was chemically synthesized. Presently it is being made by two processes.
  - Reichstein process and Two-step fermentation process

- The Reichstein process, developed in the 1930s, uses a single pre-fermentation follow by a purely chemical route. The modern two-step fermentation process, origina developed in China in the 1960s, uses additional fermentation to replace part of the la chemical stages. Both processes yield approximately 60% vitamin C from the gluco feed.

Reichstein process:

- In the Reichstein process, which is traditional and earlier, glucose is converted in sorbitol by heat treatment.
- Sorbitol is fermented and oxidised (by a microorganism Eg- *Acetobacte xylinum* or *Gluconobacter oxydans*) to form sorbose.
- This is made to react with acetone to produce di-acetone sorbose and further oxidised t produce DAKS (di-acetone keto gulonic acid).
- Then by dissolving it in a mixture organic solvent the structure of DAKS is rearrange with the help of a catalyst to form vitamin C. In the next step, by recrystallization vitamin C is purified.
- **Glucose** by heat treatment --> **Sorbitol** by fermentation --> **sorbose** by chemical proces --> **Diacetone sorbose** by chemical process --> **DAKS** by chemical process --> raw **vitamin C** by recrystallisation --> **purified vitamin C**
- This process was devised by Nobel Prize winner Tadeus Reichstein and his colleagues in 1933 while working in the laboratory of the ETH in Zürich.

The reaction steps are:

i. Hydrogenation of D-glucose to D-sorbitol, an organic reaction with nickel as a catalyst under high temperature and high pressure.

ii. Microbial oxidation or fermentation of sorbital to L-sorbose with acetobacter with pH 4-6 and 30 °C.

iii. protection of the 4 hydroxyl groups in sorbose by formation of the acetal with acetone and an acid to Diacetone-L-sorbose (2,3:4,6–Diisopropyliden–α–L–sorbose)

iv. Organic oxidation with potassium permanganate followed by heating with water gives the 2-Keto-L-gulonic acid

v. The final step is a ring-closing step or gamma lactonization with removal of water .

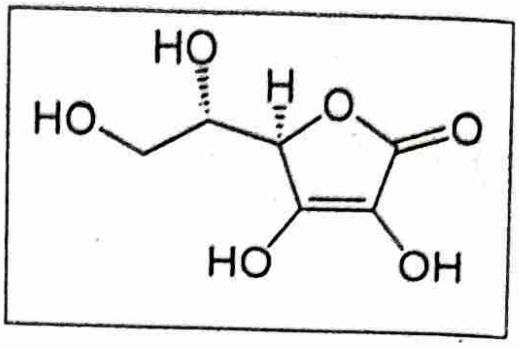
vi. Intermediate 5 can also be prepared directly from 3 with oxygen and platinum

Two-step fermentation process:

- In the two stage fermentation process developed in China, a second stage fermentation of sorbose produces KGA and this like DAKS by the same chemical process is converted into crude vitamin C and then purified.
- The later process is cost effective and more realization of ascorbic acid is achieved. Much of the world supplies of vitamin C are from China.
- **glucose** by heat treatment --> **sorbitol** by fermentation --> **sorbose** by fermentation --> **KGA** by chemical process --> **raw vitamin C** by recrystallisation --> **purified vitamin C**
- The raw material used in the synthesis of L-ascorbic acid is L-sorbose. L-sorbose, in turn, is obtained from D-sorbitol by fermentation reaction.
- The bacterium used for bringing about bioconversion is *Acetobacter suboxydans.*
- A requirement of very high level of aeration of the fermenting solution is a specific characteristic of the fermentation.
- L-sorbose has been produced by fermentation since 1934
- Vertical fermentors used for this fermentation process range from 5,000-10,000 gal capacity. They are designed so that large volumes of air may efficiently be supplied. Therefore, fermentors are equipped with a turbo mixer-type agitator and baffles along the side walls to ensure maximal turbulence of the solution.
- There is also an air inlet underneath the agitator. Air inlets may be open pipe ends, small holes or jets through which the air passes at high linear velocity or porous stones.
- The velocity jets or porous stones produce very small bubbles. Thus, this result in the maximum liquid-gas interface for oxygen transfer. Therefore, bioconversion of sorbitol to sorbose occurs at faster rates.
- At the most, fermentors may be operated at atmospheric pressure or with a head pressure up to 30 psi.
- In commercial operations with high velocity jets and 30 lb head pressure, the maximal rate of fermentation may be obtained with an air to solution volume ratio of 1:3.
- It is necessary to have provision for cooling the fermentor along with the fermenting material, since considerable heat is liberated during the fermentative reaction.
- The cooling may be achieved by circulating cold water either through coils, or preferably through jackets.

Production medium:

D-Sorbitol	200-250 gm
Cornsteep liquor	5 ml
Tap water	1 liter

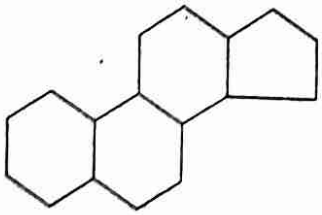


Antifoam is added as required.

- With higher sorbitol concentrations, the fermentation is slower in the beginning.
- The complete fermentation cycle takes 25-30 hours. To obtain the shortest fermentation, a solution containing 100-150 gms of sorbitol per liter is inoculated with a culture of A.suboxydans amounting to 3% of the total volume and latter enough concentrated sorbitol syrup is added to give the final sorbitol concentration.
- Analytical yields are about 95-98%. Recovery yields should be about 85%.

3.3 STEROIDS PRODUCTION:

- Steroids are a class of nonsaponifiable lipids.
- The common structural feature of steroids is that their molecules contain the following ring skeleton.

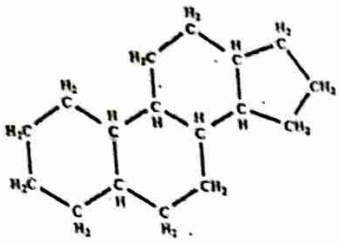


- From the chemical standpoint, hormones can generally be divided into three types: proteins (e.g., FSH and LH), peptides (e.g., GnRH and ACTH), and steroids (e.g., progesterone and estradiol).
- The basic units of proteins and peptides are amino acids, whereas steroids contain the gonane structure.
- Proteins are relatively large molecules and are water soluble, whereas steroids are small molecules and are generally soluble in organic solvents. Peptides are somewhere in between these two classes of hormones with respect to molecular weight and solubility.

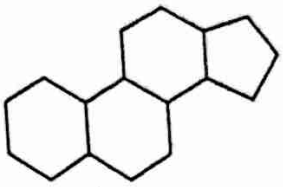
BASIC STRUCTURE AND CLASSIFICATION OF STEROIDS:

- The basic structure of steroids, gonane (cyclopentanoperhydrophenanthrene), has 17 carbons which are arranged as three six-member carbon rings to which a five-member carbon ring is attached (Fig. 1):

- Each carbon has one or two hydrogens attached.

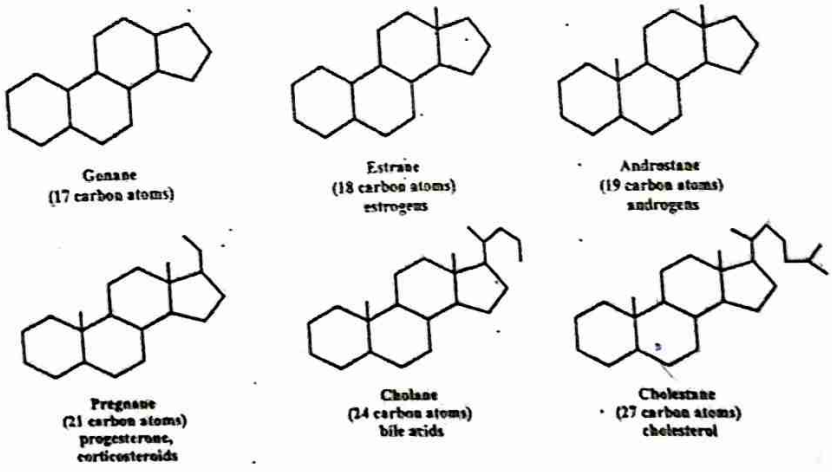


gonane



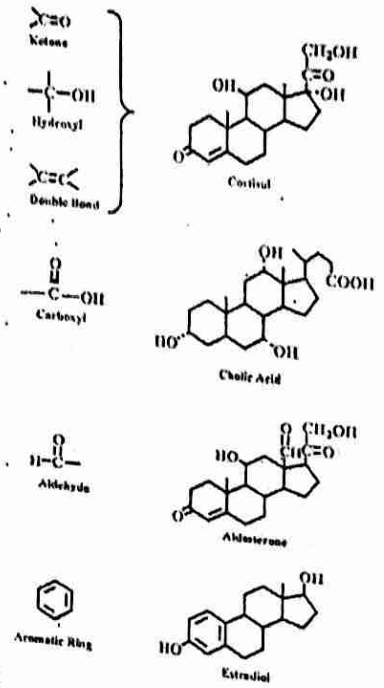
gonane

- Steroids can be divided into different groups of parent compounds, based on the number of carbons that they contain (Fig. 2).
- In addition to gonanes, which consist of 17 carbons, estranes consist of 18 carbons (C18 steroids) and include estrogens.
- Androstanes have 19 carbons (C19 steroids) and include androgens.
- Pregnanes contain 21 carbons (C21 steroids) and include progesterone and corticosteroids.



- Cholanes have 24 carbons and include bile acids, and cholestanes have 27 carbons and include cholesterol as well as cholesterol-like compounds. The compounds in this group are also referred to as sterols.
- In each group of parent steroids, compounds differ in their characteristics because of the presence of different functional groups on the molecules.

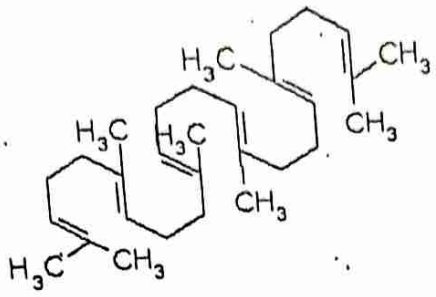
- Common functional groups include the ketone group, hydroxyl group, and double bond, as shown in the chemical structure of the cortisol molecule in Figure 3.
- Other functional groups include the carboxyl and aldehyde groups, which are present in the molecules of bile acids and aldosterone, respectively (see Fig. 3).
- An important characteristic of the C18 steroids is the presence of an aromatic ring that is found in estrogens (e.g., estradiol) (see Fig. 3).



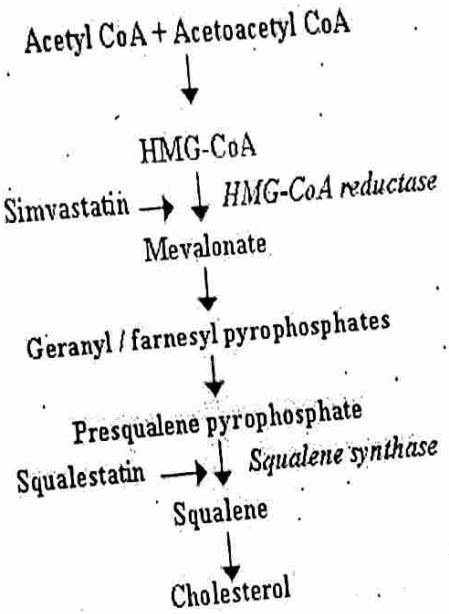
Uses of steroids:

- Steroids have a variety of uses in the human body, including, but not limited to: controlling meiosis, carbohydrate metabolism, fat storage, muscle growth, immune function and nerve cell membrane chemistry.
- Steroids can be separated into three main groups: gonadal compounds, glucocorticoids and mineralcorticoids. This distinction ***depends on the site of synthesis*** of the steroid.
- The gonadal variety are mainly synthesized in the ***gonads*** (sex gland), as is suggested by the name, while the glucocorticoids (eg cortisol, cortisone) and mineralcorticoids (eg aldosterone) are synthesized in the ***adrenal cortex***.
- Steroids are synthesized in the body from ***squalene***, a complex linear aromatic molecule shown below:

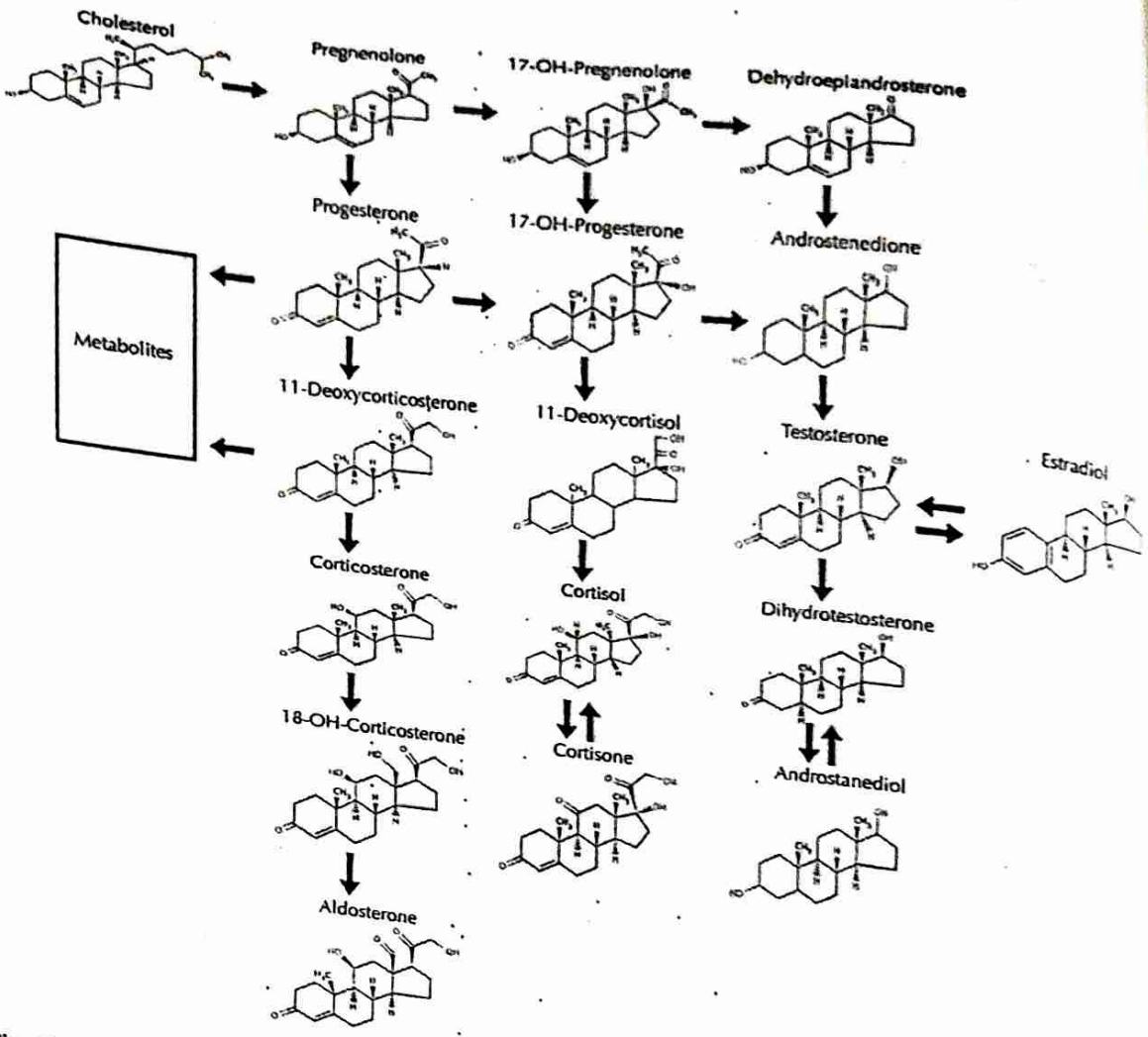
Squalene:



Cholesterol synthesis:

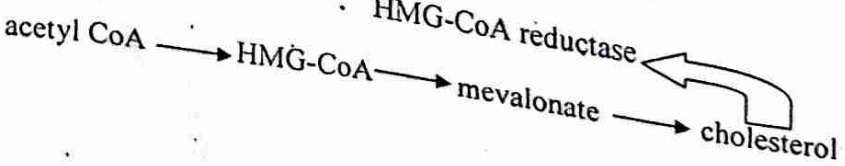


- Squalene is cyclized to form **cholesterol**. In the first few steps of this chemical pathway, squalene is converted to **squalene epoxide.**
- It is converted to **Lanosterol.** This reaction is catalyzed by the enzyme ***cyclase.*** The lanosterol is then downgraded, with three methyl groups being removed, and **cholesterol** is formed.
- Steroids are all cholesterol derivatives.



The Role of Cholesterol in Steroid Synthesis:

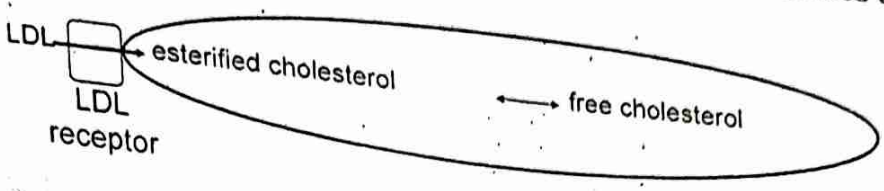
- The ***first enzymatic step*** in the production of ANY steroid hormone begins with enzymatic modification of cholesterol.
- Cholesterol can be made within the cell from acetyl CoA (de novo synthesis).
- This is a multistep process, involving many enzymatic reactions.
- A key rate-limiting enzyme is HMG-CoA reductase.
- There is negative feedback regulation of HMG-CoA reductase activity by cholesterol, so that high intracellular cholesterol inhibits de novo synthesis.



- Cholesterol is also taken up by the cell in the form of low density lipoprotein (LDL).

- LDL is a complex composed of cholesterol, phospholipids, triglycerides, and proteins (proteins and phospholipids make LDL soluble in blood).

- LDL is taken into cells via LDL receptors, and broken down into esterified cholesterol, and then free cholesterol:



Steroid Hormones:

- Steroid hormones: produced in the adrenal cortex, testis, ovary, and some peripheral tissues (adipose tissue, the brain)
- All steroid hormones share a typical (but not identical) ring structure.
- All steroid hormones are derived from cholesterol and differ only in the ring structure and side chains attached to it. All steroid hormones are lipid soluble.

Types and functions of steroid hormones:

- **Glucocorticoids**; cortisol is the major representative in most mammals. Their function is carbohydrate regulation.
- **Mineralocorticoids**; aldosterone being most prominent. Their function is mineral balance
- **Androgens** such as testosterone. Their function is reproductive functions.
- **Estrogens,** including estradiol and estrone
- **Progestogens** (also known a progestins) such as progesterone. Their function is reproductive functions.